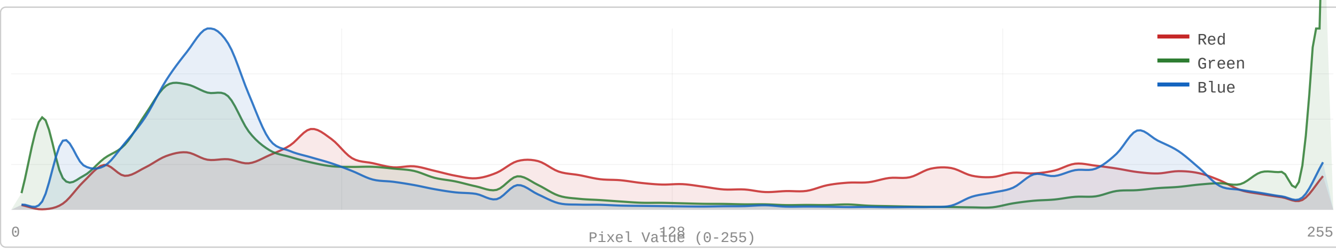
Decorrelation Gap: KODIM15

768x512 | 24-bit RGB | PNG (lossless)

Portrait of Man in Suit | Strongly one-dimensional (PC1 = 94.46%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

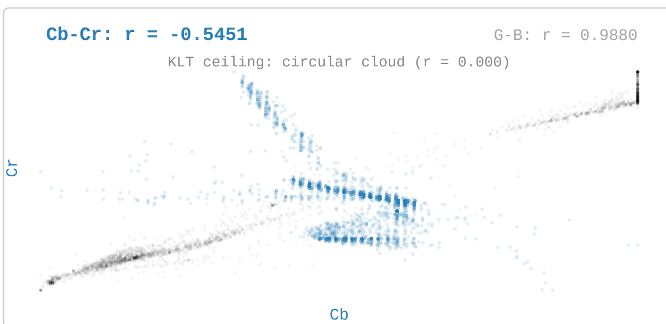
	R	G	B
R	**1.0000**	0.8602	0.8562
G	0.8602	**1.0000**	0.9880
B	0.8562	0.9880	**1.0000**

Cb-Cr:
-0.545
(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	**1.0000**	-0.0802	-0.5175
Cb	-0.0802	**1.0000**	-0.5451
Cr	-0.5175	-0.5451	**1.0000**

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.860191	-0.080225	0.0802
R-B → Y-Cr	0.856245	-0.517541	0.5175
G-B → Cb-Cr	0.987986	-0.545135	0.5451
Avg |r|	0.901474	0.380967	0.3810

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

57.7% Moderate

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

2.3%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	40.99 dB
G	46.51 dB
B	43.97 dB
Overall	43.25 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.4709	-0.6573	-0.5884	19133.44
PC2	-0.8821	0.3431	0.3227	1031.59
PC3	0.0103	-0.6710	0.7414	90.16

8. PCA Derived Metrics

Condition Number

212.23

PC1/PC2: 18.55 | PC2/PC3: 11.44

Blue Channel Independence

2.3%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.66)

9. Redundancy Profile

Classification: Strongly one-dimensional | CN: 212.23 | PC1: 94.5%

RGB Correlation: Avg |r| = 0.901474 | R-G: 0.8602 | R-B: 0.8562 | G-B: 0.9880

YCbCr Residual: Avg |r| = 0.380967 | Y-Cb: -0.0802 | Y-Cr: -0.5175 | Cb-Cr: -0.5451

Efficiency: 57.7% of KLT ceiling | Gap: 0.3810 avg |r| remaining

Blue Channel: 2.3% independent | 4:2:0 PSNR: 43.25 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.